

Shiheng Zhang
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Seattle, WA
[Google Scholar](#) | [GitHub](#)

Research Interests

Reliable low-rank computation for multiscale PDEs; dynamical low-rank approximation; geometric time integration; structure-preserving and asymptotic-preserving methods; kinetic equations.

Education

Purdue University, West Lafayette, IN 09/2019 – 12/2023
Ph.D. in Computational and Applied Mathematics
Advisor: Jie Shen
Thesis: Energy-Dissipative Methods in Numerical Analysis, Optimization and Deep Neural Networks for Gradient Flows and Wasserstein Gradient Flows

Southern University of Science and Technology, Shenzhen, China 09/2015 – 06/2019
B.S. in Computational and Applied Mathematics
Advisors: Jiang Yang and Bingsheng He

Research Experience

University of Washington, Department of Applied Mathematics, Seattle, WA 01/2024 – Present
Pearson Fellow and Acting Instructor; research with Jingwei Hu

- Dynamical low-rank algorithms for kinetic equations and related PDEs.
- Structure-preserving schemes and scientific machine learning for large-scale computation.

Awards and Honors

- Boeing Research Award, University of Washington, 2025
- T.T. Moh Graduate Scholarship, Purdue University, 2019
- Summa Cum Laude Graduate, Southern University of Science and Technology, 2019
- First Prize, Chinese Mathematics Competitions (Guangdong), Chinese Mathematical Society, 2017

Publications

Full publication list: [Google Scholar](#)

Conference Papers

- C1. Zhang, S. *Ordinal Geometry Complements Reconstruction: Diagnosing Planning with Compressed Value Functions* in *Advances in Neural Information Processing Systems* Poster (2026).

Journal Articles

- J1. Zhang, S. & Hu, J. On the stability of the low-rank projector-splitting integrator for hyperbolic and parabolic equations. *Journal of Scientific Computing* **109**, 47 (2026).
- J2. Zhang, S. & Hu, J. A separable and asymptotic-preserving dynamical low-rank method for the Vlasov–Poisson–Fokker–Planck System. *ESAIM: M2AN* **60**, 1715–1739 (2026).
- J3. Zhang, S., Shen, J. & Hu, J. SAV-based entropy-dissipative schemes for a class of kinetic equations. *SIAM Journal on Scientific Computing* (2025).
- J4. Einkemmer, L., Hu, J. & Zhang, S. Asymptotic-Preserving Dynamical Low-Rank Method for the Stiff Nonlinear Boltzmann Equation. *Journal of Computational Physics* (2025).
- J5. Zhang, S. & Shen, J. Structure preserving schemes for a class of Wasserstein gradient flows. *Communications on Applied Mathematics and Computation* (2025).
- J6. Zhang, S., Zhang, J., Shen, J. & Lin, G. A Relaxed Vector Auxiliary Variable Algorithm for Unconstrained Optimization Problems. *SIAM Journal on Scientific Computing* (2025).
- J7. Zhang, J., Zhang, S., Shen, J. & Lin, G. Energy-dissipative evolutionary deep operator neural networks. *Journal of Computational Physics* (2024).

Preprints

- R1. Zhang, S. & Hu, J. *Robust High-Order Projector-Splitting Integrators* arXiv:2608.17157.
- R2. Zhang, S. *A Counterexample to Robust Second-Order Convergence of the Strang Projector-Splitting Integrator* arXiv:2608.12893.
- R3. Zhang, S. *Guidance Breaks the Fitted Operator: A Terminal-Fitted Repair for Classifier-Free Guidance* arXiv:2607.07665.
- R4. Zhang, S. *Asymptotic Preservation and Uniform Accuracy of Diffusion and Flow-Matching Samplers* arXiv:2607.04113.
- R5. Zhang, S. & Shen, J. *SAV Schemes with Decomposition-Induced Pullback Corrections for Gradient Flows* arXiv:2606.18551.
- R6. Zhang, S. *Planner-Admissible Graph-PDE Value Extensions for Sparse Goal-Conditioned Planning* arXiv:2605.19185.
- R7. Zhang, J., Zhang, S. & Lin, G. *Low-Rank Adaptation of Evolutionary Deep Neural Networks for Efficient Learning of Time-Dependent PDEs* arXiv:2509.16395.

Talks and Posters

Talks

- T1. Zhang, S. *On the stability of the low-rank projector-splitting integrator for parabolic equations* Talk, CHaRMNET Annual Meeting, University of Washington, December 2025. 2025.
- T2. Zhang, S. *Structure preserving schemes for a class of Wasserstein gradient flows* Invited minisymposium talk, SIAM Annual Meeting, Spokane, WA, July 2024. 2024.
- T3. Zhang, S. *A Relaxed Vector Auxiliary Variable Algorithm for Unconstrained Optimization Problems* Invited minisymposium talk, SIAM Conference on Optimization, Seattle, WA, June 2023. 2023.

Posters

- P1. Zhang, S. *A Separable and Asymptotic-Preserving Dynamical Low-Rank Method for the Vlasov–Poisson–Fokker–Planck System* Poster, IPAM, UCLA, Los Angeles, CA, March 2026. 2026.
- P2. Zhang, S. *SAV-based entropy-dissipative schemes for a class of kinetic equations* Poster, ICERM, Providence, RI, July 2025. 2025.
- P3. Zhang, S. *Asymptotic-Preserving Dynamical Low-Rank Method for the Stiff Nonlinear Boltzmann Equation* Poster, CHaRMNET Annual Meeting, Virginia Tech, December 2024. 2024.
- P4. Zhang, S. *SAV-based entropy-dissipative schemes for a class of kinetic equations* Poster, NSF Computational Mathematics PI Meeting, University of Washington, July 2024. 2024.

Teaching

Instructor, University of Washington

- AMATH 351 A and B: Introduction to Differential Equations and Applications – Spring 2026
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2026
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2025
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2025
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2024

Teaching Assistant, Purdue University

Courses: MA 265: Linear Algebra; MA 266: Ordinary Differential Equations; MA 261: Multivariate Calculus; MA 162: Calculus II; MA 161: Calculus I

Technical Skills

- **Programming:** Python (PyTorch, JAX, TensorFlow), Julia, C++, MATLAB, Git, SQL
- **Research Areas:** Numerical optimization, dynamical low-rank methods, scientific machine learning, numerical analysis for PDEs, structure-preserving schemes